

Qatar-2b

R-1831

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_150421 · created 2026-08-03T05:50:29+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457133.871933 +/- 8.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.012
Transit depth (Rp/Rs)^2	2.51 +/- 0.39 [%]
Orbital Inclination (inc)	89.52 +/- 1.25
Airmass coefficient 1 (a1)	1.002 +/- 0.017
Airmass coefficient 2 (a2)	-0.0011 +/- 0.0077
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	#2 - [118, 206]
Optimal Aperture	2.81
Optimal Annulus	10.6
Transit Duration (day)	0.0762 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.012 vs 0.16327 (deviation 0.33σ)
Duration fit vs published	0.0762 d vs 0.0767 d (deviation 0.34σ)
Mid-time O–C	0.3 min
Depth SNR	12.4
Residual scatter	1.69 %

Light curve

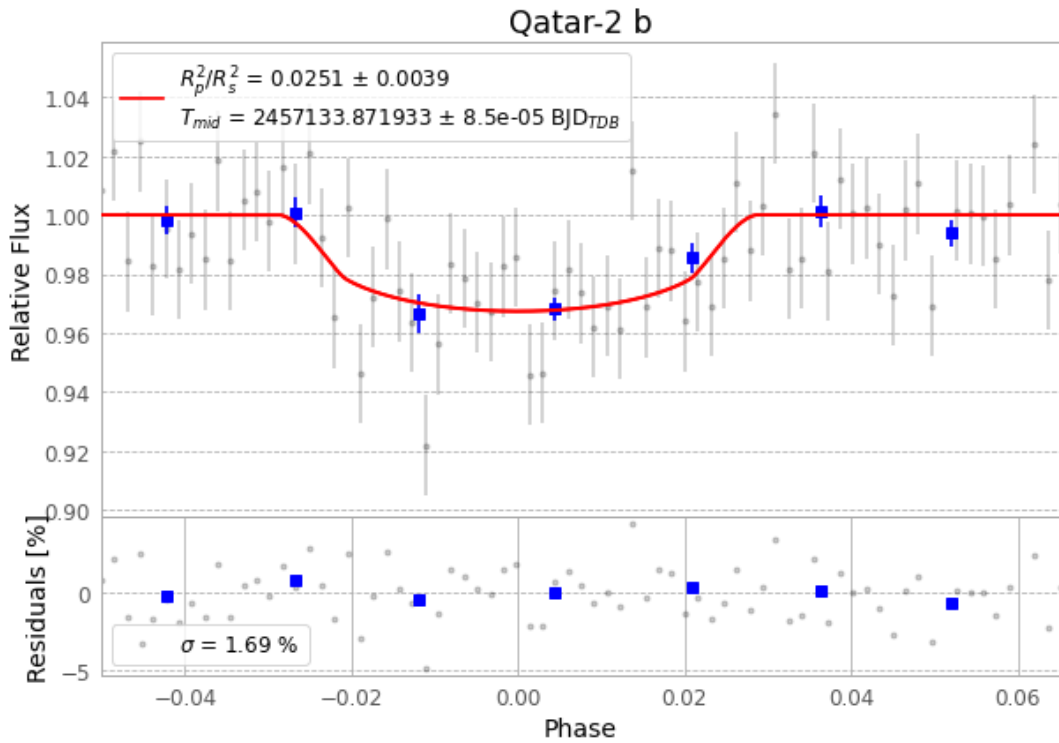


Figure 1 — FinalLightCurve_Qatar-2 b_2015-04-21.png (SHA-256 4af9ade57978b1e3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [22, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 90f2022138ce626d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), Qatar-2150421070912.FITS → Qatar-2150421105112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-150421.log (fca1f446b898...), FinalLightCurve_Qatar-2 b_2015-04-21.png (4af9ade57978...), FinalLightCurve_Qatar-2 b_2015-04-21.csv (a5e945664fef...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:50Z.